

Optimizing neural network architecture using genetic algorithms, particle swarm optimizations, and simulated annealing

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Overview

- 1 Introduction
- 2 Algorithms
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Introduction - Problem Description

- **Given** a dataset with n data points $d^{(1)}, d^{(2)}, \dots, d^{(n)} \in \mathbb{R}^m$, where $n, m \in \mathbb{N}$
Find a model $\mathcal{M}$

$$f() :=$$

- **Example:**
- **Motivation** (Optimization of NNA):

Introduction

- Non-linear cost function $f() =$
- Discrete / Continuous solution space $D_s = \{\}$
- TBD
- Solution space is represented by soggy cat
- **Naive Approach:** Brute-Force Algorithm with run time $\mathcal{O}(2^x)$ (x still has to be determined)

Introduction - Project Objectives

- Investigate how well the **Genetic Algorithm** (normal and enhanced) and **Particle Swarm Optimization** perform when applied to the problem.
- TBD
- TBD
- Assess the performance of these algorithms by comparing them to the **Brute-Force** and **Monte-Carlo** approach.

Algorithms - Brute Force and Monte-Carlo

Algorithms - Simulated Annealing

Algorithms - (Enhanced) Genetic Algorithm

Algorithms - Particle Swarm Optimization

Parametric Optimization

Parametric Optimization

Parametric Optimization - Genetic Algorithm

Results

Conclusion and Discussion

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